Laboratory 07: Logistic Regression
Project results and model evaluation

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May 8, 2026

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Chapter 1

Standard logistic regression analysis

In this section, we analyze the performance of a standard (non-prior-weighted) Logistic Regression model on the project dataset. The goal is to evaluate the impact of the regularization parameter λ on both the discriminative capability (minDCF) and the calibration (actDCF) of the model for a target application with $\pi_T = 0.1$.

1.1 Experimental Results

The model was trained using 13 values of λ logarithmically spaced between 10^{-4} and 10^2 [cite: 4-17]. [cite_start]For each value, we computed the minDCF and actDCF, after converting the raw scores into Log-Likelihood Ratios (LLRs) by removing the training set's empirical prior log-odds. The numerical results are summarized in Table 1.1.

Table 1.1: Standard Logistic Regression Performance ($\pi_T = 0.1$)

λ	minDCF	actDCF
1.0000×10^{-4}	0.3640	0.4021
1.0000×10^{-3}	0.3650	0.4130
1.0000×10^{-2}	0.3611	0.4568
1.0000×10^{-1}	0.3641	0.8522
1.0000×10^{0}	0.3640	1.0000
1.0000×10^{1}	0.3630	1.0000
1.0000×10^{2}	0.3620	1.0000

1.2 Performance Analysis

What do you observe? Based on the results in Table 1.1 and Figure 1.1, we observe that the minDCF remains remarkably stable around a value of 0.36 across the entire range of λ. In contrast, the actDCF is relatively close to the minDCF only for very small values of λ (e.g., 0.4021 at $\lambda = 10^{-4}$). As λ increases, the actDCF rises sharply, eventually reaching a value of 1.0 for $\lambda \geq 1$, which corresponds to the cost of a non-informative system.

Can you see significant differences for the different values of lambda? There are no significant differences in terms of the model's potential discriminatory power (minDCF), as the fluctuations are minimal. However, there is a very significant difference in terms of calibration (actDCF). The model is reasonably well-calibrated for low λ, but becomes completely mis-calibrated as regularization increases, showing that λ has a much stronger impact on the reliability of the scores than on their ranking ability.

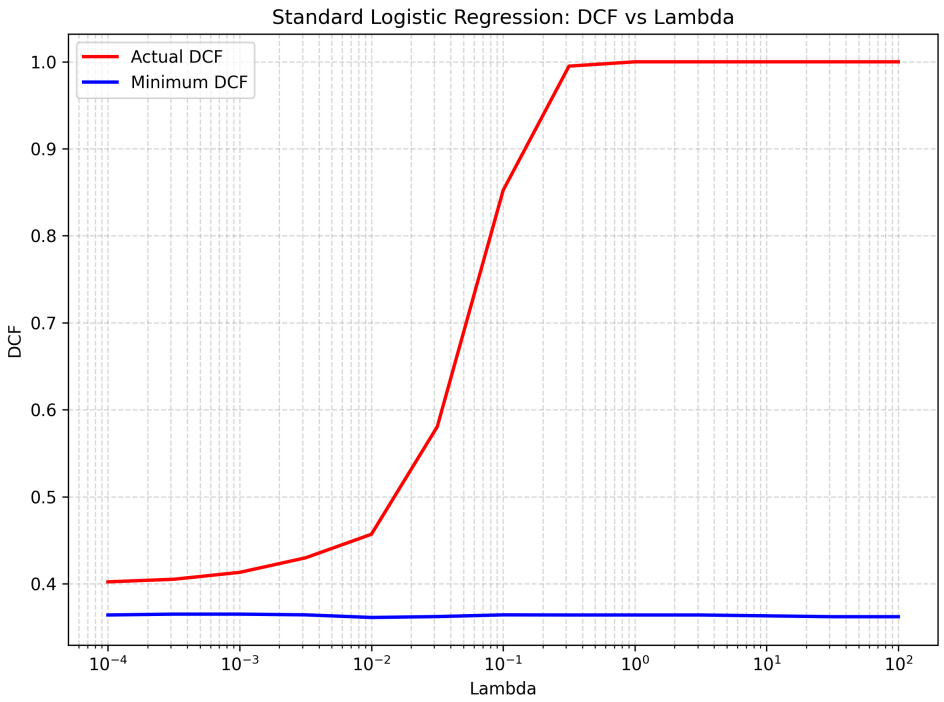


Figure 1.1: Standard Logistic Regression: DCF vs λ ($\pi_T = 0.1$)

How does the regularization coefficient affects the two metrics? The regularization coefficient λ affects the two metrics in fundamentally different ways:

- **Effect on minDCF:** The impact is negligible here. This is because the training dataset is sufficiently large, meaning the model is not prone to overfitting in this linear configuration. Therefore, penalizing the magnitude of the weights does not change the relative ordering of the validation samples significantly.
- **Effect on actDCF:** Regularization has a severe negative impact on actDCF. Theoretically, Logistic Regression minimizes the cross-entropy to produce scores that behave like LLRs. As $\lambda \to \infty$, the penalty forces the weights $w \to 0$. When weights are extremely small, the scores $s(x) = w^T x + b$ become nearly constant, losing their probabilistic meaning and their alignment with the application's required threshold. This leads to the massive "calibration gap" observed in the plots.

Chapter 2

Reduced dataset analysis

In this section, we analyze the behavior of the Logistic Regression model when training data is scarce. By keeping only 1 out of every 50 samples from the original training set, we reduce the number of observations available for parameter estimation. This setup allows us to highlight the role of regularization as a tool to prevent overfitting in low-data regimes.

2.1 Experimental results

The model was trained on the reduced dataset $DTR_{reduced}, LTR_{reduced}$ using the same range of λ values as the previous analysis. The validation was still performed on the full validation set to ensure a reliable estimate of the generalization error.

Table 2.1: Reduced dataset logistic regression results ($\pi_T = 0.1$)

λ	minDCF	actDCF
1.0000×10^{-4}	0.4466	0.9780
1.0000×10^{-3}	0.4487	0.7466
1.0000×10^{-2}	0.4407	0.4652
1.0000×10^{-1}	0.3988	0.7164
1.0000×10^{0}	0.3874	1.0000
1.0000×10^{1}	0.3783	1.0000
1.0000×10^{2}	0.3783	1.0000

2.2 Analysis and Discussion

What do you observe?

In this reduced data scenario, the behavior of the metrics changes significantly compared to the full-dataset case. The minDCF is no longer flat; it starts at a higher value (≈ 0.44) for low λ and gradually improves (decreases) as λ increases, reaching its best value around $\lambda = 10$.

More importantly, the actDCF curve now exhibits a clear "U-shape." For very small values of λ, the actDCF is extremely high (≈ 0.98), nearly matching the cost of a non-informative system. It reaches a minimum at an intermediate value ($\lambda = 10^{-2}$) before rising again to 1.0 as regularization becomes too strong.

Can you explain the results in this case? Remember that lower values of the regularizer imply larger risk of overfitting, while higher values of the regularizer reduce overfitting, but may lead to underfitting and to scores that lose their probabilistic interpretation.

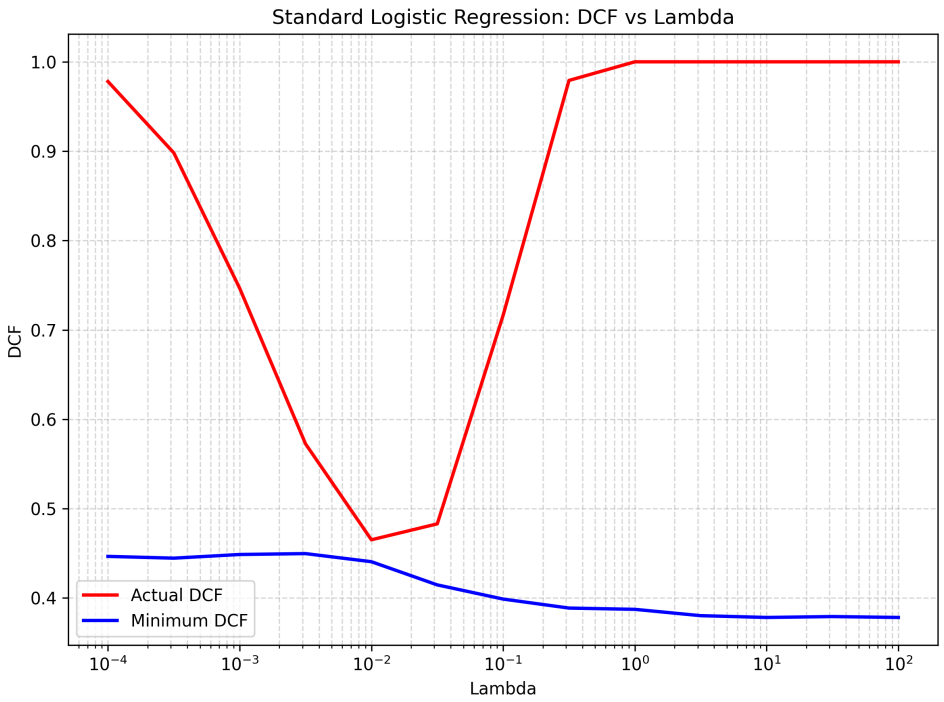


Figure 2.1: Reduced Dataset: DCF vs λ ($\pi_T = 0.1$)

The results can be explained by the relationship between model complexity, sample size, and regularization:

- **Low regularization ($\lambda < 10^{-2}$): Overfitting.** With only 1/50th of the data, the model has very few samples to define the decision boundary. For small λ, the optimizer is free to choose weights that "perfectly" fit these few points, including their specific noise. This leads to extremely overconfident scores. While the ranking might be somewhat correct (minDCF ≈ 0.44), the scores are poorly calibrated, resulting in an actDCF near 1.0. The model has high variance and fails to generalize.

- **Optimal regularization ($\lambda \approx 10^{-2}$): Generalization.** At this intermediate range, the regularization term effectively penalizes large weight values. This prevents the model from fitting the noise of the small training set too closely. Theoretically, regularization reduces the effective capacity of the model, bringing the empirical risk closer to the true risk. This results in the best-calibrated scores and the lowest actDCF (0.4652).

- **High regularization ($\lambda > 10^{-1}$): Underfitting and calibration Loss.** As λ increases further, the penalty term dominates the cross-entropy loss. The weights w are pushed toward zero, and the model becomes too simple to capture the underlying data distribution (underfitting). Furthermore, as the weights shrink, the scores $s(x) = w^T x + b$ are compressed toward a constant value. These "shrunken" scores no longer represent valid Log-Likelihood Ratios, losing their probabilistic interpretation. Consequently, the system can no longer make informative decisions at the application threshold $\pi_T = 0.1$, and the actDCF returns to 1.0.

Interestingly, the minDCF improves for high λ because the reduction in weight magnitude can sometimes lead to a more stable ranking of samples in a low-data regime, even if the absolute values of the scores are useless for direct thresholding.

2.3 Theoretical conclusion

In conclusion, when data is abundant, the model is naturally regularized by the volume of information. However, when data is scarce, explicit regularization via λ is mandatory to bridge the gap between minimum and actual performance by controlling the trade-off between bias (underfitting) and variance (overfitting).

Chapter 3

Prior-weighted logistic regression analysis

In this section, we analyze the performance of the Prior-weighted logistic regression model. Unlike the standard model, this version explicitly incorporates the target application prior $\pi_T = 0.1$ into the objective function during the training phase.

3.1 Experimental results

The model was trained on the full project dataset using 13 values of λ logarithmically spaced between 10^{-4} and 10^{2}. To obtain the Log-Likelihood Ratios (LLRs), we subtracted the log-odds of the target prior used during training ($\pi_T = 0.1$).

Table 3.1: Prior-weighted logistic regression results ($\pi_T = 0.1$)

λ	minDCF	actDCF
1.0000×10^{-4}	0.3721	0.4071
1.0000×10^{-3}	0.3699	0.4129
1.0000×10^{-2}	0.3630	0.4487
1.0000×10^{-1}	0.3648	0.9147
1.0000×10^{0}	0.3640	1.0000
1.0000×10^{1}	0.3630	1.0000
3.1623×10^{1}	0.3620	1.0000

3.2 Analysis and discussion

Are there significant differences for this task?

No, there are no significant differences in terms of performance between the standard and the prior-weighted models for this specific task. Comparing the best minimum DCF obtained by the standard model (0.3611) with the best minimum DCF of the prior-weighted model (0.3620), the values are nearly identical. Both models exhibit the same trend: a stable minDCF across the lambda range and a degrading actDCF that reaches 1.0 as regularization increases.

The lack of difference is motivated by the fact that the project dataset is sufficiently large. In high-data regimes, the model is able to learn the underlying class distributions effectively regardless of whether the classes are weighted in the objective function or not, provided the empirical prior of the training set is not extremely skewed compared to the target application.

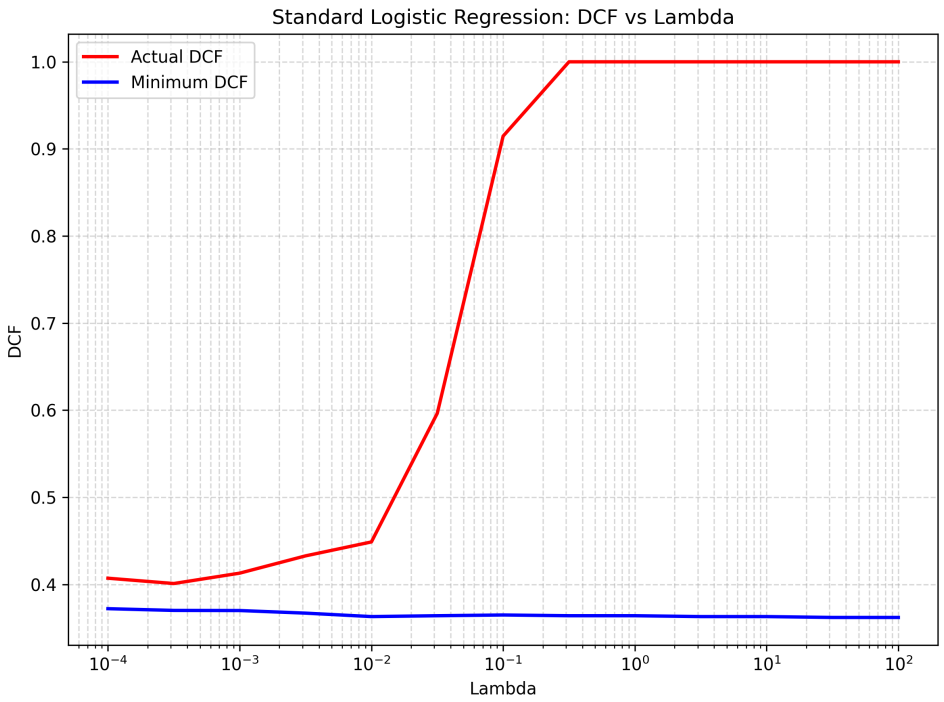


Figure 3.1: Prior-weighted logistic regression: DCF vs λ ($\pi_T = 0.1$)

Are there advantages using the prior-weighted model for our application (remember that the prior-weighted model requires that we know the target prior when we build the model)?

In this specific case, the prior-weighted model does not offer a numerical advantage, but it provides a theoretical one. The primary advantage of the prior-weighted model is its ability to "focus" the training on a specific operating point. By incorporating π_T into the loss function, we are telling the optimizer to penalize errors on the minority class more heavily (in our case, the impostors if we consider the target prior of 0.1).

However, this approach has two main drawbacks:

- **Requirement of prior knowledge:** We must know the target application prior at training time. If the application changes (e.g., if we need to evaluate for $\pi_T = 0.5$), the prior-weighted model trained for 0.1 will be poorly calibrated and may yield sub-optimal results.
- **Calibration sensitivity:** As seen in the results, even with the correct prior weighted into the training, the actDCF still diverges from the minDCF at high lambda values. This confirms that regularization still compresses the scores, leading to a loss of probabilistic meaning that the prior weighting alone cannot fix.

Theoretically, the prior-weighted model is most advantageous when the training set is highly imbalanced or unrepresentative of the real-world

Chapter 4

Quadratic logistic regression analysis

In this section, we move beyond linear decision boundaries by applying a quadratic feature expansion to the project dataset. By mapping the original features into a higher-dimensional space that includes all pairwise products of the features, we allow the Logistic Regression model to learn non-linear boundaries in the original feature space.

4.1 Experimental results

The model was trained on the full expanded dataset using a standard (non-prior-weighted) objective function. We evaluated the performance across the same range of λ values for the primary application $\pi_T = 0.1$.

Table 4.1: Quadratic logistic regression results ($\pi_T = 0.1$)

λ	minDCF	actDCF
1.0000×10^{-4}	0.2602	0.2768
1.0000×10^{-3}	0.2587	0.2765
3.1623×10^{-3}	0.2527	0.2771
1.0000×10^{-2}	0.2487	0.3464
3.1623×10^{-2}	0.2436	0.4972
1.0000×10^{-1}	0.2466	0.7520
1.0000×10^{0}	0.2840	1.0000
1.0000×10^{2}	0.3263	1.0000

4.2 Analysis and discussion

What do you observe?

The most striking observation is the significant improvement in discriminatory power compared to the linear models. The minimum DCF drops from approximately 0.36 in the linear case to a best value of 0.2436 in the quadratic case. This indicates that the relationship between the features and the classes is inherently non-linear, and the quadratic expansion successfully captures this complexity.

Furthermore, we observe that the minDCF is no longer flat as it was in the linear full-dataset analysis; it shows a distinct downward trend as λ increases toward 10^{-2} before rising again. The actDCF remains very close to the minDCF for small values of λ ($\leq 10^{-3}$), showing excellent calibration initially, but it degrades much faster than the linear model as regularization increases.

In this case is regularization effective?

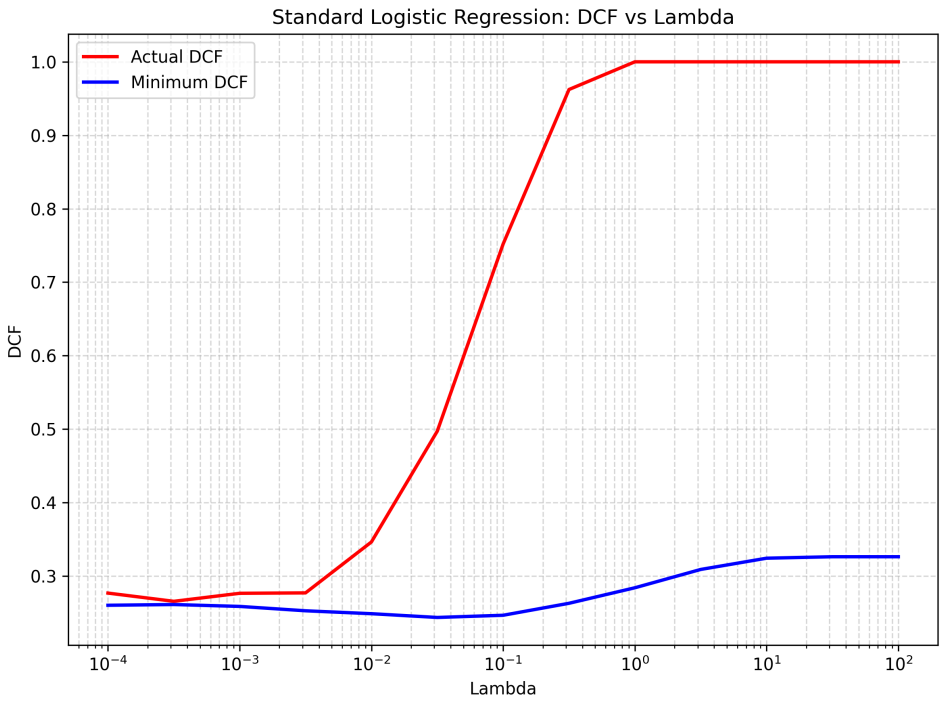


Figure 4.1: Quadratic logistic regression: DCF vs λ ($\pi_T = 0.1$)

Yes, in the quadratic case, regularization is highly effective and necessary. In the linear analysis on the full dataset, the number of parameters was small relative to the number of samples, making regularization almost irrelevant for the minDCF. However, the quadratic expansion significantly increases the number of parameters (the dimensionality of the weight vector w).

With more parameters, the model has a higher capacity, which introduces a greater risk of overfitting the training data. We can see regularization at work by looking at the minDCF: it improves as we increase λ from 10^{-4} up to 3.1623×10^{-2}. This shows that penalizing the weights helps the model generalize better to the validation set by preventing it from fitting noise in the expanded feature space.

How does it affect the two metrics?

The regularization coefficient λ affects the metrics as follows:

- **Effect on minDCF:** Regularization manages the bias-variance tradeoff. For very low λ, the model overfits (higher minDCF due to variance). As λ increases toward 3×10^{-2}, the minDCF reaches its optimum because the regularization prevents over-fitting without losing the signal. Beyond this point, the model begins to underfit (higher minDCF due to bias), as the penalty prevents the weights from capturing the actual patterns in the data.

- **Effect on actDCF:** For low λ, the calibration is quite good because the quadratic model is complex enough to accurately model the log-posterior probabilities. However, as λ increases beyond the optimal point, the actDCF explodes. Theoretically, this happens because the regularization pushes the weights toward zero, compressing the scores $s(x) = w^T \phi(x) + b$. These compressed scores lose their alignment with the required LLR threshold for $\pi_T = 0.1$, resulting in a massive increase in the actual cost, eventually reaching the default cost of 1.0.

4.3 Theoretical motivation

The superiority of the quadratic model stems from the fact that it can model covariance differences between classes even when using a discriminative approach. While a linear model is roughly equivalent to a Tied Covariance Gaussian model (assuming a linear boundary), the quadratic expansion allows the Logistic Regression model to simulate the behavior of an MVG model with different covariance matrices for each class. The results confirm that the flexibility provided by the expanded feature space $\phi(x) = [vec(xx^T), x]$ is crucial for this specific dataset.

Chapter 5

Final model comparison and conclusions

In this final section, we consolidate the results obtained across all experimental setups, including the generative Gaussian models from previous laboratories and the discriminative Logistic Regression models developed in this laboratory. The comparison is based on the minimum DCF for the primary application $\pi_T = 0.1$, as this metric represents the optimal discriminatory potential of each model regardless of its current calibration.

5.1 Summary of results

The following table summarizes the best performance achieved by each model type. For Logistic Regression models, the results corresponding to the optimal regularization parameter λ are reported.

Table 5.1: Overall model comparison ($\pi_T = 0.1$)

Model	Optimal λ	minDCF	actDCF
MVG (Full Covariance)	-	0.2629	0.3051
Naive Bayes	-	0.2570	0.3022
Tied MVG	-	0.3628	0.4061
Standard Logistic Regression	10^{-2}	0.3611	0.4568
Prior-Weighted Logistic Regression	3.16×10^{1}	0.3620	1.0000
Quadratic Logistic Regression	3.16×10^{-2}	**0.2436**	0.4972

5.2 Analysis and comparative discussion

Which model(s) achieve(s) the best results?

The best overall result is achieved by the **Quadratic Logistic Regression** model, which reaches a minimum DCF of 0.2436. Among the generative models, **Naive Bayes** (0.2570) and the **Multivariate Gaussian (MVG)** model (0.2629) also perform significantly better than any of the linear models. The linear Logistic Regression models and the Tied MVG model consistently show higher (worse) minDCF values, plateauing around 0.36.

What kind of separation rules or distribution assumptions characterize this / these model(s)?

The top-performing models share a common characteristic: they allow for **non-linear decision boundaries**.

- **Quadratic logistic regression:** This is a discriminative model that does not make explicit assumptions about the distribution of the features. However, by expanding the feature space to include quadratic terms, it learns a quadratic decision boundary. This allows the model to capture differences in the "spread" or orientation (covariances) of the two classes.
- **MVG and Naive Bayes:** These are generative models. MVG assumes that each class follows a Gaussian distribution with its own specific mean and full covariance matrix. Naive Bayes is a simplified version that assumes diagonal covariance matrices (feature independence). These models naturally result in quadratic decision boundaries (Quadratic Discriminant Analysis) when the covariance matrices of the classes differ.
- **Linear models (Standard LR, Tied MVG):** These models assume, either explicitly or implicitly, that the classes can be separated by a simple hyperplane. Tied MVG assumes that both classes share the same covariance matrix, which mathematically restricts the decision boundary to be linear. The poor performance of these models indicates that a linear rule is insufficient for this dataset.

How are the results related to the characteristics of the dataset features?
The results provide strong evidence regarding the statistical structure of the project dataset:

1. **Covariance mismatch:** The fact that Quadratic LR and MVG significantly outperform Tied MVG and Linear LR strongly suggests that the two classes (authentic and impostor) have **different covariance structures**. In other words, the way features correlate with each other in the "authentic" class is different from how they correlate in the "impostor" class. A linear model cannot capture this "second-order" information.
2. **Quadratic benefit:** The Quadratic Logistic Regression outperforms the MVG model (0.2436 vs 0.2629). This suggests that while the classes might be "Gaussian-like," they do not perfectly follow a Gaussian distribution. As a discriminative model, Logistic Regression is more robust to modeling errors because it focuses directly on the decision boundary rather than trying to model the entire density of the features.
3. **Feature interaction:** The high performance of the quadratic expansion indicates that interactions between different features (cross-terms) carry significant discriminative information that is lost when considering features only in their linear form.

5.3 Final considerations on calibration

While Quadratic Logistic Regression provides the best discriminatory potential (lowest minDCF), it also shows a significant **calibration gap** ($actDCF - minDCF = 0.2536$). This reinforces the laboratory's conclusion that complex models, especially those with many parameters or high regularization, require post-processing calibration to ensure that their scores can be reliably used as Log-Likelihood Ratios in a real-world application. Currently, the generative models (MVG and Naive Bayes) appear more "naturally" calibrated for this specific application.